

Developer Mock Test Results

Student: Ch Vineeshaa

Name:	Ch Vineeshaa	Student ID:	113
Date:	2026-03-05 12:36:25	Performance:	■■ Needs Improvement
Overall Score:	9.0/25 (36.0%)	Test ID:	f0880d81-3d62-4554...
Warnings:	0/3	Status:	Completed ■

Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	2/10	20.0%	■■ Needs Work
■ MCQ/Theory	5/10	50.0%	■ Pass
■ Coding	2.0/5	40.0%	■■ Needs Work

Detailed Question Review

■ APTITUDE SECTION (2/10 - 20.0%)

■ Q1. If 5 men can build a wall in 10 days, how many men are needed to build the same wall in 5 days?

Your Answer: 12

Correct Answer: 10

■ To find the number of men needed to build the wall in 5 days, we can use the formula: $\text{men} \times \text{days} = \text{constant}$. Given that 5 men can build the wall in 10 days, we have $5 \times 10 = 50$ man-days. To build the same wall in 5 days, we need $50 \text{ man-days} / 5 \text{ days} = 10$ men. The user's answer of 12 is incorrect, likely due to a miscalculation of the proportionality between the number of men and the number of days. The correct calculation shows that 10 men are needed to build the wall in 5 days.

■ Q2. If it is true that all A are B and some B are not C, what can be concluded about A and C?

Your Answer: No A are C

Correct Answer: Some A are C

■ The correct answer is "Some A are C" because if all A are B and some B are not C, it doesn't necessarily mean that A and C are unrelated or that no A are C. The user's likely mistake is assuming that since some B are not C, A (being a subset of B) cannot be C, which is a flawed conclusion. The correct interpretation is that since all A are B, and B has both C and non-C elements, A could potentially overlap with either or both, making "Some A are C" a plausible conclusion.

■ Q3. A product is on sale for 30% off its original price of \$150. How much will you pay for the product?

Your Answer: \$100

Correct Answer: \$105

■ To find the sale price, calculate 30% of the original price of \$150 and subtract it from \$150. The key calculation step is: $\$150 \times 0.30 = \45 , then $\$150 - \$45 = \$105$. The user's answer of \$100 is likely incorrect because they may have miscalculated the discount amount or subtracted it incorrectly from the original price.

■ Q4. If it is true that all cats are animals and all dogs are animals, what can be concluded about cats and dogs?

Your Answer: They are different species

Correct Answer: They are both animals

■ The correct answer is "They are both animals" because the given statements establish a common category between cats and dogs, which is that they are both classified as animals. The user's answer, "They are different species," is likely incorrect because the statements do not provide information about their species classification, only that they share the category of being animals. The user's mistake is likely assuming that the statements imply a comparison of their species, when in fact they only establish a shared characteristic.

■ Q5. What is the average of 2, 4, 6, 8, 10?

Your Answer: 6

■ Well done!

■ Q6. A product is on sale for 25% off its original price of \$100. How much will you pay for the product?

Your Answer: \$85

Correct Answer: \$75

■ To find the sale price, calculate 25% of the original price (\$100) and subtract it from the original price. The key calculation step is: $\$100 - (0.25 * \$100) = \$100 - \$25 = \$75$. The user likely made a mistake by not correctly calculating the discount amount, leading them to choose \$85 instead of the correct answer, \$75.

■ Q7. If it is true that all cars are vehicles and some vehicles are bicycles, what can be concluded?

Your Answer: No cars are bicycles

Correct Answer: Some vehicles are cars

■ The correct answer is "Some vehicles are cars" because we know that all cars are vehicles, which implies that cars are a subset of vehicles. This means that at least some vehicles are indeed cars. The user's answer, "No cars are bicycles", is incorrect because the statement "some vehicles are bicycles" does not provide enough information to conclude that cars and bicycles are mutually exclusive.

■ Q8. If 3 men can build a wall in 8 days, how many men are needed to build the same wall in 4 days?

Your Answer: 6

■ Excellent! Right answer.

■ Q9. A mixture is made of 3 parts coffee and 4 parts milk. What is the ratio of coffee to milk?

Your Answer: 4:3

Correct Answer: 3:4

■ The correct answer is 3:4 because the mixture is made of 3 parts coffee and 4 parts milk, directly giving us the ratio of coffee to milk. The user's answer, 4:3, is likely a mistake due to reversing the given parts of coffee and milk. The key calculation step is simply comparing the given parts: coffee (3 parts) to milk (4 parts), resulting in a ratio of 3:4.

■ Q10. A product is on sale for 35% off its original price of \$120. How much will you pay for the product?

Your Answer: \$80

Correct Answer: \$78

■ To find the sale price, calculate 35% of the original price (\$120) and subtract it from the original price. The key calculation step is: $\$120 * 0.35 = \42 , then $\$120 - \$42 = \$78$. The user's answer of \$80 is likely due to a calculation error, as the correct calculation yields \$78.

■ MCQ SECTION (5/10 - 50.0%)

■ Q1. What is an interface in Java?

Your Answer:

A abstract class that cannot be instantiated

■ Correct! Good understanding.

■ Q2. What is a livelock in Java?

Your Answer: A situation where two or more threads are able to access a shared resource

Correct Answer: A situation where two or more threads are unable to access a shared resource

■ The correct answer is A) A situation where two or more threads are unable to access a shared resource, because a livelock occurs when multiple threads are constantly trying to access a shared resource, but due to repeated failures and retries, none of them can actually access it. The user's answer, B) A situation where two or more threads are able to access a shared resource, is incorrect because it describes the opposite of a livelock, where threads are successfully accessing the resource. The user's likely mistake is misunderstanding the concept of livelock, which is similar to a deadlock, but with the added aspect of threads continuously trying to access the resource, rather than being completely blocked.

■ Q3. What is abstraction in Java?

Your Answer: The concept of hiding all internal details and showing no information

Correct Answer: The concept of showing only necessary information and hiding internal details

■ The correct answer is A) The concept of showing only necessary information and hiding internal details, because abstraction in Java is a fundamental concept that allows developers to expose only the necessary features of an object or system while concealing its internal implementation details. This helps to reduce complexity, improve modularity, and increase flexibility. The user's answer, C) The concept of hiding all internal details and showing no information, is incorrect because abstraction is not about hiding everything, but rather about showing only what is necessary, which is a key distinction.

■ Q4. What is exception handling in Java?

Your Answer: The process of handling runtime errors

■ Excellent! Right answer.

■ Q5. What is logging in Java?

Your Answer: The process of recording all types of information in a program

Correct Answer: The process of recording events and errors in a program

■ The correct answer, "The process of recording events and errors in a program", is right because logging in Java refers to the practice of tracking and recording significant events, errors, and exceptions that occur during the execution of a program. This helps developers diagnose and debug issues, as well as monitor the program's performance. The user's answer, "The process of recording all types of information in a program", is incorrect because while logging can record various types of information, its primary focus is on events and errors, not all types of information.

■ Q6. How should Java code be organized?

Your Answer: Using methods and variables

Correct Answer: Using classes and packages

■ The correct answer, "Using classes and packages", is right because Java code is typically organized into classes, which are the basic building blocks of Java programs, and packages, which are used to group related classes together. This organization helps to promote modularity, reusability, and maintainability of the code. The user's answer, "Using methods and variables", is incorrect because while methods and variables are essential components of Java code, they are not the primary means of organizing code, but rather are used within classes and packages to implement specific functionality.

■ Q7. What are the types of errors in Java?

Your Answer: All of the above

■ Excellent! Right answer.

■ Q8. What is synchronization in Java?

Your Answer: The process of coordinating access to shared resources

■ Excellent! Right answer.

■ Q9. How are variables declared in Java?

Your Answer: Using the "let" keyword

Correct Answer: Using the data type

■ The correct answer is "Using the data type" because in Java, variables are declared by specifying the data type of the variable, followed by the variable name. This is in contrast to some other programming languages that use keywords like "var", "let", or "const" for variable declaration. The user's likely mistake is confusing Java with JavaScript, which uses the "let" keyword for variable declaration, highlighting the importance of distinguishing between different programming languages.

■ Q10. When should comments be written in Java?

Your Answer: For complex logic

■ Correct! Well done.

■ CODING SECTION (2.0/5 - 40.0%)

■ Q1. Write a Java program that reads a string from stdin and counts the frequency of each word, using a lambda expression to sort the results. ****Input For...**

Your Answer:

```
import java.util.*;

public class WordFrequency {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        String input = sc.nextLine();
        String[] words = input.split("\\s+");

        Map<String, Integer> freqMap = new LinkedHashMap<>();

        for (String word : words) {
            freqMap.put(word, freqMap.getOrDefault(word, 0) + 1);
        }

        freqMap.entrySet()
            .stream()
            .sorted((a, b) -> {
                int freqCompare = b.getValue().compareTo(a.getValue());
                if (freqCompare == 0) {
                    return 0; // keep original order
                }
                return freqCompare;
            })
            .forEach(entry -> System.out.print(entry.getKey() + ":" + entry.getValue() + " "));
    }
}
```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on achieving an overall "Accepted" status, keep up the good work!

■ Q2. Write a Java program that reads an array of integers from stdin and removes duplicates. ****Input Format:** - Line 1: The number of elements in the arr...**

Your Answer:

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        solution();
    }

    public static void solution() {
        Scanner sc = new Scanner(System.in);
```

```

        int n = sc.nextInt();
        LinkedHashSet<Integer> set = new LinkedHashSet<>();

        for (int i = 0; i < n; i++) {
            set.add(sc.nextInt());
        }

        for (int num : set) {
            System.out.print(num + " ");
        }
    }
}

```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on writing accurate and effective code that meets the requirements.

■ **Q3. Write a Java program that reads a string from stdin and prints its reverse. ****Input Format:**** - Line 1: A string ****Output Format:**** - The reversed s...**

Your Answer:

No answer (Skipped)

Correct Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        String reversed = new StringBuilder(input).reverse().toString();
        System.out.println(reversed);
        scanner.close();
    }
}

```

■ There is no test case failure to analyze, as no test case results or student code were provided. Without this information, it's impossible to determine what went wrong. Please provide the test case results and student code to receive accurate feedback.

■ **Q4. Write a Java program that reads a string from stdin and counts the frequency of each word. ****Input Format:**** - Line 1: A string ****Output Format:**** -...**

Your Answer:

No answer (Skipped)

Correct Answer:

```

import java.util.HashMap;
import java.util.Map;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        String[] words = input.split("\\s+");

        Map<String, Integer> wordFrequencies = new HashMap<>();

        for (String word : words) {
            wordFrequencies.put(word, wordFrequencies.getOrDefault(word, 0) + 1);
        }
    }
}

```

```

        for (Map.Entry<String, Integer> entry : wordFrequencies.entrySet()) {
            System.out.print(entry.getKey() + ":" + entry.getValue() + " ");
        }

        scanner.close();
    }
}

```

■ There is no test case failure to analyze, as no test case results or student code were provided. Without this information, it's impossible to determine what went wrong. Please provide the test case results and student code to receive accurate feedback.

■ **Q5. Write a Java program that reads an integer from stdin and checks if it is a perfect cube. **Input Format:** - Line 1: An integer **Output Format:** ...**

Your Answer:

No answer (Skipped)

Correct Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int number = scanner.nextInt();
        double cubeRoot = Math.cbrt(number);
        if (cubeRoot == Math.round(cubeRoot)) {
            System.out.println("Perfect Cube");
        } else {
            System.out.println("Not Perfect Cube");
        }
        scanner.close();
    }
}

```

■ There is no code submitted to analyze, and no test case results are provided, so it's impossible to determine what went wrong. The student hasn't submitted any code for review. No issues can be identified without the actual code or test case results.

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.